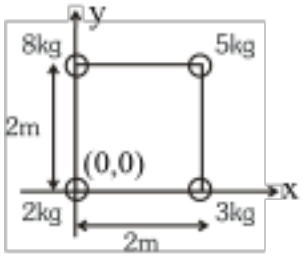
PHYSICS

1)

Position of the centre of mass of system is :-



(1) $\left(\frac{8}{9}, \frac{13}{9}\right)$

(2) $\left(\frac{13}{9}, \frac{8}{9}\right)$

(3) $\left(\frac{15}{9}, \frac{8}{9}\right)$

(4) $\left(\frac{8}{9}, \frac{15}{9}\right)$

2) A projectile is thrown at a speed of 100 m/sec. at an angle of 37° above the horizontal. At the highest point the projectile breaks into two parts of mass ratio 1 : 3, the smaller coming to rest. Find the speed of second piece.

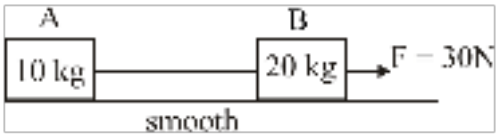
(1) 320/3 m/s

(2) 310/3 m/s

(3) 800/3 m/s

(4) 1120/3 m/s

3) Two blocks A and B are connected by a massless string. A force of 30N is applied on block B. The distance travelled by centre of mass of system in 2 second starting from rest is -



(1) 1m

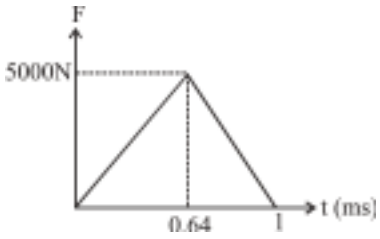
(2) 2m

(3) 3m

(4) none

4) Graph of force of impact 'F' vs time 't' for short duration of collision of two bodies is as shown.

Find coefficient of restitution for given graph.



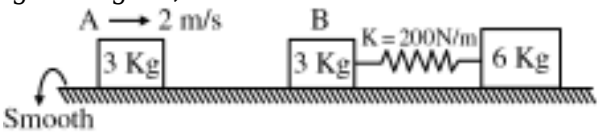
(1) 0.56

(2) 1

(3) 0.64

(4) 0.26

5) In the given figure, if collision between block A and B is elastic, maximum compression in spring will be:-



(1) 10 cm

(2) 20 cm

(3) 30 cm

(4) None

6) A road of width 20 m forms an arc of radius 15 m, its outer edge is 2 m higher than its inner edge. For what speed the rod is banked?

(1) $\sqrt{10}$ m/s

(2) $\sqrt{14.7}$ m/s

(3) $\sqrt{9.8}$ m/s

(4) None of these

7) A particle moves in a circle of radius 10 cm with constant speed and time period π s. The acceleration of the particle is :-

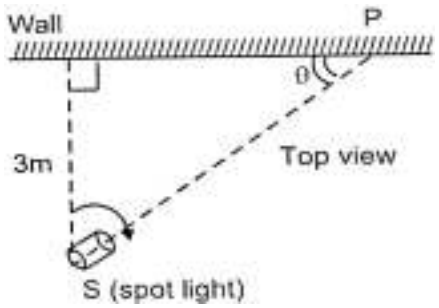
(1) 10 cm/s^2

(2) 20 cm/s^2

(3) 40 cm/s^2

(4) 50 cm/s^2

8) A spot light S rotates in a horizontal plane with a constant angular velocity of 0.1 rad/s. The spot P of light moves along the wall at a distance 3m. What is the velocity of the spot P when θ = 45º ?



(1) 0.6 m/s

(2) 0.5 m/s

(3) 0.4 m/s

(4) 0.3 m/s

9) A simple pendulum oscillates in a vertical plane. When it passes through the mean position, the tension in the string is 3 times the weight of the pendulum bob. What is the maximum angular displacement of the pendulum of the string with respect to the vertical :

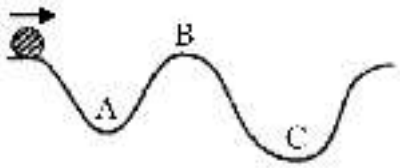
(1) 30º

(2) 45º

(3) 60º

(4) 90º

10) A body moves along a path as shown with a constant speed at all points. The normal reaction of the path on the body-



(1) maximum at A

(2) maximum at B

(3) minimum at C

(4) same at A, B and C

11) A planet of mass 'm' is moving in an elliptical orbit about the sun with time period 'T'. If 'A' be the area of orbit, then its angular momentum would be :-

(1) $\frac{2mA}{T}$

(2) mAT

(3) $\frac{mA}{2T}$

(4) 2mAT

12) For energy of satellite, match the columns (symbols have their respective meaning)

	Column-I		**Column-II**

(i)	Kinetic energy	(p)	$\frac{L^2}{2mr^2}$
(ii)	Potential energy	(q)	$-\frac{L^2}{mr^2}$
(iii)	Total energy	(r)	$-\frac{L^2}{2mr^2}$

(1) i-q ; ii-p, ; iii-r

(2) i-p ; ii-r ; iii-q

(3) i-p ; ii-q ; iii-r

(4) i-r ; ii-q ; iii-p

13) ***Assertion :*** If rotation of earth about its own axis is suddenly stopped then acceleration due to gravity will increase at all places on the earth (except poles).

Reason : At height h from the surface of earth, acceleration due to gravity is $g_h = g\left(1-\frac{2h}{R_e}\right)$. (If $h << R_e$) [$R_e \rightarrow$ radius of earth]

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion.**

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

14) The gravitational field in a region is given by $\vec{I} = 10(\hat{i}+\hat{j})$ N/kg. The work done by gravitational field to shift a particle of mass 2 kg from position (0,0) to (5, 4) will be :-

(1) 100 J

(2) 180 J

(3) 80 J

(4) 20 J

15) Two bodies, each of mass M, are kept fixed with a separation 2L. A particle of mass m is projected from the mid-point of the line joining their centres, perpendicular to the line. The gravitational constant is G. The correct statement(s) is (are) :-

(a) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $4\sqrt{\frac{GM}{L}}$

(b) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $2\sqrt{\frac{GM}{L}}$

(c) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $\sqrt{\frac{2GM}{L}}$

(d) The energy of the mass m remains constant

(1) a, b

(2) b, d

(3) a, c

(4) a, d

16) The gravitational potential energy of a body of mass m at the earth's surface is $-mgR_e$. Its gravitational potential energy at a height R_e from the earth's surface will be (Here R_e is the radius of the earth)

(1) $-2mgR_e$

(2) $2mgR_e$

(3) $\frac{mgR_e}{2}$

(4) $\frac{-mgR_e}{2}$

17) Two identical satellites are at height R and 7R from earth surface, the ratio of their kinetic energies will be :-

(1) 2

(2) 1

(3) 4

(4) Infinite

18) The weight of a body at a distance r (r > R) from the centre of earth is W. The weight at a distance 2r from the centre of earth is :- (R → Radius of earth)

(1) $\frac{W}{3}$

(2) $\frac{W}{4}$

(3) $\frac{W}{2}$

(4) $\frac{W}{6}$

19) If mass of a planet is M and radius is x, then the work to be done to slowly take a mass m from surface of planet to a height 4x is :-

(1) $\frac{4GMm}{5x}$

(2) $\frac{2GMm}{5x}$

(3) $\frac{5GMm}{x}$

(4) $\frac{4GMm}{3x}$

20) A long capillary is dipped in beaker containing water. Water rises in capillary upto some height h. Match the statements in list-I with most appropriate effects on water level mentioned in list-II

	List-I		List-II
A.	Soap solution is added to water	p	h decreases
B.	Arrangement taken in a freely falling lift	q	h increases
C.	Arrangement taken in a lift accelerating uniformly upward	r	h remains same
D.	Arrangement is taken in a lift accelerating uniformly downward	s	water will rise upto complete height of capillary
		t	water level in capillary goes below the outside level

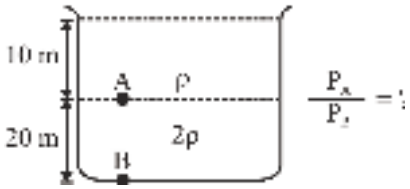
(1) A - p, B - t, C - q, D - p

(2) A - t, B - q, C - s, D - p

(3) A - p, B - s, C - p, D - q

(4) A - q, B - p, C - s, D - q

21) From the given figure ratio of pressure at A and B will be (ρ = 1000 kg/m^3) [$p_0 = 10^5$pa]

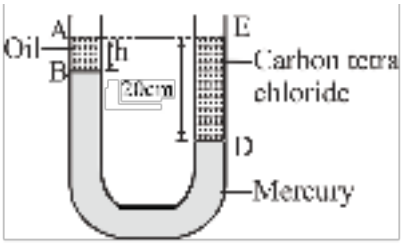


(1) 1/2

(2) 1/4

(3) 1/3

(4) 2/3



22) Calculate the value of h in U-tube shown in the following figure.
Density of oil = 0.9 g/cm^3
Density of carbon tetra chloride = 1.6 g/cm^3
Density of mercury = 13.6 g/cm^3

(1) 18.9 cm

(2) 20.9 cm

(3) 16.9 cm

(4) 14.9 cm

23) A water tank resting on the floor has two small holes vertically one above the other. The holes are h_1 cm and h_2 cm above the floor. How high does water stand in the tank if the jets from the holes

hits the floor at the same point ?

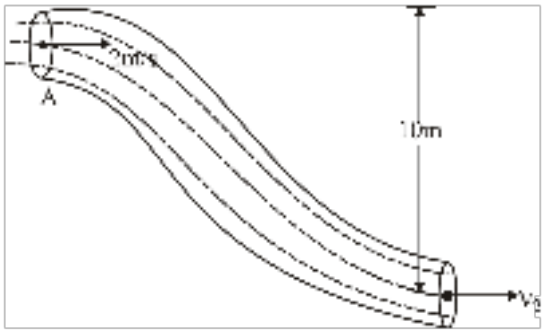
(1) $(h_1 + h_2)$

(2) $(h_2 - h_1)$

(3) $\frac{h_1^2 + h_2^2}{2}$

(4) $\frac{h_2^2 - h_1^2}{2}$

24) Water gradually descends 10 m in height as shown in figure. If velocity at point A is 2m/s, then find V_2 in this tube of uniform cross-section.



(1) 10 m/s

(2) 2 m/s

(3) $\sqrt{204}$ m/s

(4) 4 m/s

25) A liquid is flowing in a cylindrical pipe of internal diameter 4cm with a velocity of 5m/s . If this tube is joined with another tube of internal diameter 2cm then the velocity of flow of liquid in the smaller tube will be (in ms^{-1})

(1) 10

(2) 40

(3) 5

(4) 20

26) Young's modulus of rubber is 10^4 N/m^2 and area of cross-section is 2 cm^2. If force of 2×10^5 dynes is applied along its length, then length of wire becomes how much times of its initial length L:-
(Assume stress $\propto$ strain)

(1) three times

(2) four times

(3) two times

(4) No change in length

27) A metal block is experiencing an atmospheric pressure of 1×10^5 N/m^2. When the same block is placed in a vacuum chamber, the fractional change in its volume is (the bulk modulus of metal is 1.25×10^{11} N/m^2)

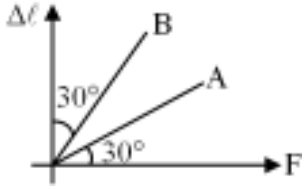
(1) 4×10^{-7}

(2) 2×10^{-7}

(3) 8×10^{-7}

(4) 1×10^{-7}

28) Elongation ($\Delta\ell$) Vs Force (F) graph of two wires A and B is shown in the figure. Both wires are made of same material and have same volume Then which option is incorrect (where Y is Young's modulus is length of wire, A is area of cross section of wire and r is radius of wire.)



(1) $y_A = y_B$

(2) $_B > _A$

(3) $A_B < A_A$

(4) $r_B > r_A$

29) When the load on a wire is increased from 3 kg wt to 5 kg wt the elongation increases from 0.61 mm to 1.02 mm. The required work done during the extension of the wire is :-

(1) 16×10^{-3} J

(2) 8×10^{-2} J

(3) 20×10^{-2} J

(4) 11×10^{-3} J

30) The bulk modulus for an incompressible liquid is :-

(1) zero

(2) unity

(3) infinity

(4) between 0 and 1

31) Terminal velocity of iron ball of radius 1 mm in glycerine is v_1 and v_2 is the terminal velocity of lead ball of radius 2 mm in glycerine then $v_1 : v_2$ is :
[$\rho_{glycerine}$ = 13.6 g/cc, σ_{Fe} = 7.6 g/cc,
σ_{Pb} = 11.6 g/cc]

(1) 0.47

(2) 0.58

(3) 0.75

(4) 0.21

32) A liquid is flowing in a horizontal uniform capillary tube under a constant pressure difference P. The value of pressure for which the rate of flow of the liquid is doubled when the radius and length

both are doubled is:

(1) P

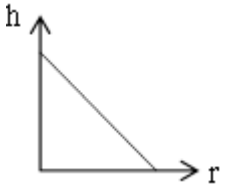
(2) $\frac{3P}{4}$

(3) $\frac{P}{2}$

(4) $\frac{P}{4}$

33) The correct curve between the height or depression h of liquid in a capillary tube and its radius is :

(1)



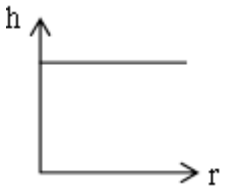
(2)



(3)



(4)



34)

If the excess pressure inside a soap bubble is balanced by an oil column of height 2 mm, then the surface tension of soap solution will be :

(r = 1 cm and density d = 0.8 gm/cc)

(1) 4 N/m

(2) 4×10^{-2} N/m

(3) 4×10^{-4} N/m

(4) 4 dyne/m

35) A ring of radius 1.5 cm is floating horizontally on the surface of water. If this ring has to be raised up then how much additional force has to be applied to lift ring : (Surface tension of water 73×10^{-3} Newton/metre)

(1) 1.37×10^{-2} N

(2) 2.3×10^{-2} N

(3) 5.1×10^{-3} N

(4) 4×10^{-2} N

36) What is ratio of surface energy of 1 small drop and 1 large drop, if 1000 small drops combined to form 1 large drop :-

(1) 100 : 1

(2) 1000 : 1

(3) 10 : 1

(4) 1 : 100

37) A particle is moving along a straight line parallel to x-axis with constant velocity. Its angular momentum about the origin :

(1) Decreases with time

(2) Increases with time

(3) Remains constant

(4) Is zero

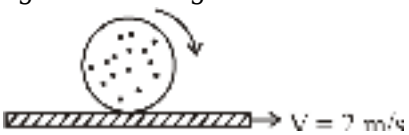
38) If the rotational kinetic energy of a body increased by 300% then determine the percentage increase in its angular momentum:

(1) Increases by 100%

(2) Decreases by 100%

(3) Increases by 50%

(4) None of these

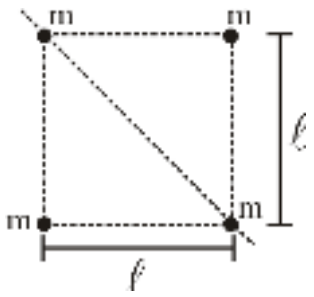
39) A uniform disc of radius 20 cm is performing pure rolling on a moving surface. If its angular velocity is 5 rad/s then find velocity of it's centre of mass:- 

(1) 1 m/s

(2) 2 m/s

(3) 3 m/s

(4) None of these

40) Find moment of inertia of system about diagonal of square :- 

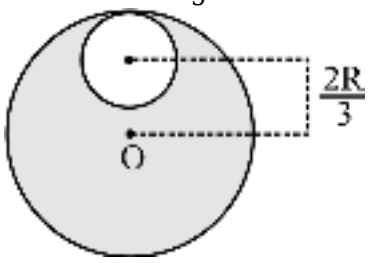
(1) $2m\square^2$

(2) $m\square^2$

(3) $\frac{m\ell^2}{2}$

(4) $\frac{m\ell^2}{\sqrt{2}}$

41) From a circular disc of radius R and mass 9M a small disc of radius $\frac{R}{3}$ is removed as shown in figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the



disc and passing through O is :

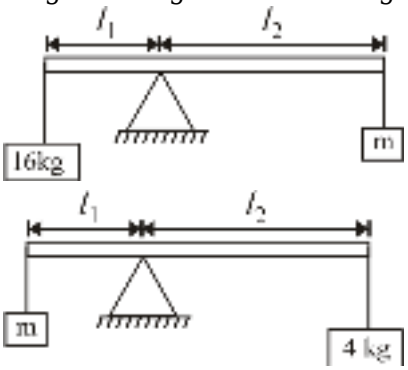
(1) $4\ MR^2$

(2) $\frac{40}{9}MR^2$

(3) $10\ MR^2$

(4) $\frac{37}{9}MR^2$

42) In an experiment with a beam balance an unknown mass m is balanced by two known masses of 16 kg and 4 kg as shown in figure.



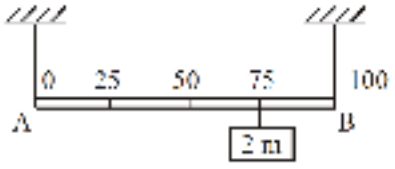
Find the value of the unknown mass m in kg.

(1) 2 kg

(2) 4 kg

(3) 8 kg

(4) 12 kg



43) Shown in the figure is rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of

mass 'm' and has another weight of mass 2 m hung at a distance of 75 cm from A. The tension in the string at A is :

(1) 2 mg

(2) 0.5 mg

(3) 0.75 mg

(4) mg

44) A ladder of length and mass m is placed against a smooth vertical wall but the ground is not smooth. Coefficient of friction between the ground and the ladder is μ. The minimum angle θ with ground at which the ladder will stay in equilibrium is :

(1) $\tan^{-1}(\mu)$

(2) $\tan^{-1}(2\mu)$

(3) $\tan^{-1}(\mu/2)$

(4) $\tan^{-1}(1/2\mu)$

45) A disc of mass 2 kg and radius 0.2 m is rotating with angular velocity 30 rad/sec. If a mass of 0.25 kg is put gently on periphery of disc then angular velocity of disc is :-

(1) 24 rad/sec

(2) 36 rad/sec

(3) 15 rad/sec

(4) 26 rad/sec

CHEMISTRY

1) Wave number $(\bar{\nu})$ for a radiation of 6000 Å wavelength is :-

(1) $6.66 \times 10^{6}\ m^{-1}$

(2) $1.66 \times 10^{6}\ m^{-1}$

(3) $1.66 \times 10^{7}\ m^{-1}$

(4) $6 \times 10^{-7}\ m^{-1}$

2) In an atom, total number of electrons having n = 4, $|m_\ell| = 1$ and $m_s = +\frac{1}{2}$ is :-

(1) 1

(2) 6

(3) 12

(4) 4

3) Match list-I with list-II :-

	List-I		**List-II** $[a_0 = 0.529Å]$

(A)	3rd orbit of Li^{2+} ion	(i)	$E_n = -13.6$ eV $r = 3a_0$
(B)	2nd orbit of Be^{3+} ion	(ii)	$E_n = -3.4$ eV $r = 8a_0$
(C)	2nd orbit of H-atom	(iii)	$E_n = -3.4$ eV $r = 4a_0$
(D)	4th orbit of He^{+} ion	(iv)	$E_n = -54.4$ eV $r = a_0$

Select the correct answer.

(1) A-i, B-iv, C-iii, D-ii

(2) A-i, B-iv, C-ii, D-iii

(3) A-iv, B-iii, C-ii, D-i

(4) A-iv, B-ii, C-iii, D-i

4) Given below are two statements :-

Statement-I :- $|\psi|^2$ at any point in an atom gives the value of probability density at that point.

Statement-II :- For 1s orbital, probability density is maximum at the nucleus and decreases sharply as distance from nucleus increases.

In light of above statements, choose the most appropriate answer :-

(1) Both statement-I and statement-II are true.

(2) Both statement-I and statement-II are false.

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false but statement-II is true .

5) If the radius of first orbit of H-atom is a_0, then de-Broglie wavelength of electron in 3rd obit of H atom is :-

(1) $2\pi a_0$

(2) $\frac{\pi a_0}{3}$

(3) $6\pi a_0$

(4) $3\pi a_0$

6) Shortest wavelength of H-atom in Lyman series is λ, then longest wavelength in Balmer series is :-

(1) λ

(2) $\frac{5}{36}\lambda$

(3) $\frac{36}{5}\lambda$

(4) $\frac{\lambda}{4}$

7) Uncertainty in the position of an electron ($m_e = 9.1 \times 10^{-31}$ kg) moving with a velocity of 300

m/sec, accurate upto 0.001%, will be : ($h = 6.63 \times 10^{-34}$ J-s)

(1) 5.76×10^{-2} m

(2) 1.92×10^{-2} m

(3) 3.84×10^{-2} m

(4) 19.2×10^{-2} m

8) Select the incorrect statement :-

(1) Principal quantum number (n) determines the size of orbital.

(2) For a given value of ℓ, m_ℓ has $(2\ell + 1)$ values.

(3) An orbital cannot hold more than 2 electrons and these two electrons should have opposite spin.

(4) For a given value of n, ℓ can have (n – 1) values from 0 to (n – 1)

9) Maximum number of spectral lines obtained in H-spectrum if electron make transition from 5th excited state to ground state.

(1) 15

(2) 6

(3) 10

(4) 5

10) Which of the following is not a limitation of Bohr's model ?

(1) It fails to explain the spectrum of H-atom.

(2) Bohr's theory was unable to explain splitting of spectral lines in presence of magnetic field.

(3) It could not explain the ability of atoms to form molecules by chemical bonds.

(4) It ignores dual behaviour of matter and also contradicts with Heisenberg uncertainty principle.

11) Given below are two statements :
Statement-I :- For a given shell, total number of allowed orbitals is equal to n.
Statement-II :- Total number of angular nodes are given by (n – 1).
In light of above statements, select the most appropriate answer.

(1) Both statement-I and statement-II are incorrect.

(2) Both statement-I and statement-II are correct.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

12) Match list-I with list-II :

	List-I (Set of quantum numbers)		**List-II (Maximum no. of electrons in an atom)**
(A)	$n = 4, m_\ell = 0$	(I)	2

(B)	n = 2, □ = 0	(II)	8
(C)	n = 3, □ = 1, $s = +\frac{1}{2}$	(III)	16
(D)	n = 4, $m_s = +\frac{1}{2}$	(IV)	3

Select the correct option.

(1) A-III, B-I, C-II, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-II, B-IV, C-III, D-I

(4) A-III, B-II, C-IV, D-I

13) Energy of 1st Balmer line of H-atom is xJ, then energy (in J) of second Balmer line of H-atom is :-

(1) x

(2) 2x

(3) 1.35x

(4) $\frac{x}{1.35}$

14)

Which of the following ionisation energy order is correct ?

(1) $Cu^{+}(3d^{10}) > Cu^{+2}(3d^{9})$

(2) $Mn^{+2}(3d^{5}) > Mn^{+3}(3d^{4})$

(3) $N^{-3}(3p^{6}) > N(2p^{3})$

(4) $Ar < Ar^{\oplus}$

15) $[Xe]4f^{14}\ 5d^{10}\ 6s^{2}$ configuration belongs to :-

(1) s-block

(2) p-block

(3) d-block

(4) f-block

16) The set of elements that show incorrect order of Z_{eff} :- (Acc. to slater)

(1) Al > Mg

(2) Na > Li

(3) K > Na

(4) Si > Al

17) Which statement is incorrect :-

(1) In alkali metals group, from top to bottom increase in size is maximum from Na to K

(2) Addition of e^- in P atom will be exothermic

(3) Ionisation energy of F is greater than its electron affinity value.

(4) Reaction is $O^-_{(g)} + S_{(g)} \rightarrow O_{(g)} + S^-_{(g)}$ endothermic

18) Correct order of electron gain enthalpy is.

(1) H < Li < Na

(2) F > Cl > Br

(3) O < Se < S

(4) Cl > O > F

19) Amongst the following the highest Ionisation energy will be of :-

(1) [Ne] $3s^2 3p^1$

(2) [Ne] $3s^2 3p^3$

(3) [Ne] $3s^2 3p^2$

(4) [Ar] $3d^{10} 4s^2 4p^3$

20) Correct order of increasing metallic character is-

(1) Si < P < Be < Mg < Na

(2) P < Si < Be < Mg < Na

(3) P < Si < Mg < Be < Na

(4) P < Si < Be < Na < Mg

21) **Assertion :** Noble gases have zero e^- gain enthalpy.
Reason : All noble gases have ($ns^2 np^6$) stable outer electronic configuration.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

22) Match the Column I with Column II and select correct answer by given codes. :-

Column I (Element types)	**Column II (Electronic configuration)**
A. Inert-gas elements	1. $(n-1)d^{1-10} ns^{1-2}$
B. Transition elements	2. ns^2 and $ns^2 np^6$
C. Inner-transition elements	3. $(n-2)f^{1-14}$ $(n-1)s^2 p^6 d^{0-1} ns^2$

(1) A – 1, B – 2, C – 3

(2) A – 2, B – 1, C – 3

(3) A – 3, B – 2, C – 1

(4) A – 2, B – 3, C – 1

23) The pair of amphoteric hydroxide is :-

(1) $Al(OH)_3$, LiOH

(2) $Be(OH)_2$, $Mg(OH)_2$

(3) $B(OH)_3$, $Be(OH)_2$

(4) $Be(OH)_2$, $Zn(OH)_2$

24) Which of the following pair has same hybridisation :-

(1) BF_3 & NF_3

(2) SF_4, $SiCl_4$

(3) ClO_4^- & ClO_2^-

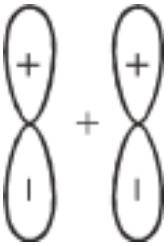
(4) CO_2 & SiO_2

25) The combination of orbital which will form a nodal plane ?

(1) 

(2) 

(3) 

(4) 

26) ***Assertion :-*** B_2 molecule is diamagnetic.
Reason :- The highest occupied molecular orbital (H.O.M.O.) is of π-type in B_2.

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

27) Match the column :-

	Column-I		**Column-II**
(a)	H_3O^+	(i)	Square pyramidal
(b)	HC ≡ CH	(ii)	Pyramidal

(c)	IF_5	(iii)	See-saw
(d)	SF_4	(iv)	Linear

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(ii), b-(iv), c-(i), d-(iii)

(3) a-(iv), b-(iii), c-(ii), d-(i)

(4) a-(iii), b-(ii), c-(iv), d-(i)

28) Which molecule is polar and non-planar?

(1) SO_2

(2) CCl_4

(3) ClF_3

(4) SF_4

29) The vapour pressure of *o*-nitrophenol at any given temperature is predicted to be–

(1) higher than that of *p*-nitrophenol

(2) lower than that of *p*-nitrophenol

(3) same as that of *p*-nitrophenol

(4) higher or lower depending upon the size of the vessel

30) In which of the following the central atom does not use sp^3 hybrid orbitals in its bonding :-

(1) BeF_3^-

(2) OH_3^+

(3) NH_2^-

(4) NF_3

31) The atomic number of three elements A, B and C are a, a + 1 and a + 2, C is an alkali metal. In a compound of A and C, the nature of bonding is :-

(1) Co-ordinate

(2) Covalent

(3) Ionic

(4) Metallic

32) **Bond Bond dissociation energy (kJ mole^{-1})**

C-I 240 → Element A
C-II 328 → Element B
C-III 276 → Element C
C-IV 485 → Element D

Elements A, B, C and D, which element has the smallest atom ?

(1) I

(2) II

(3) III

(4) IV

33) An element (X) forms compounds of the formula XCl_3, X_2O_5 and Ca_3X_2 but does not for XCl_5. Predict element (X) :

(1) B

(2) Al

(3) N

(4) P

34) Normally, Sulphur does not exist as S_2 molecule because :-

(1) it is less electronegative

(2) it is not able to form stable pπ-pπ bond

(3) it has ability to exhibit catenation

(4) of tendency to show variable oxidation states

35)

Which of the following compound have number of pπ–pπ bond equal to pπ–dπ bond.

(1) SO_3

(2) SO_2

(3) $SOCl_2$

(4) SO_2Cl_2

36) In which of the following d-orbitals are not used by central atom in hybridization ?

(1) $PF_5(s)$

(2) $PCl_5(s)$

(3) $PBr_5(s)$

(4) $XeF_6(s)$

37) Which of the following statements is incorrect ?

(1) In O_2 molecules, the energy of σ_{2pz} is more than π_{2py}

(2) C_2, O_2^{-2} and N_2 are diamagnetic

(3) He_2 molecule does not exist as the effect of bonding and anti-bonding orbitals cancel each other

(4) Among O_2^+, O_2 and O_2^- the order of bond strength is $O_2^+ > O_2 > O_2^-$

38) Which of the following order is incorrect ?

(1) Ionic character : $MCl < MCl_2 < MCl_3$

(2) Polarisibility : $F^- < Cl^- < Br^- < I^-$

(3) Polarising power : $Na^{+} < Mg^{+2} < Al^{+3}$

(4) Covalent character : LiF<LiCl<LiBr<LiI

39) Select correct order out of given options :-

(1) $BeCO_3 < BaCO_3 \rightarrow$ Covalent character

(2) BeO > SrO → Lattice energy

(3) $Be^{2+} < Li^{+} \rightarrow$ Hydration energy

(4) $Be^{2+}_{(aq.)} > Li^{+}_{(aq.)} \rightarrow$ Ionic Mobility

40) Which of the following is least soluble in acetone

(1) AgCl

(2) NaCl

(3) CCl_4

(4) LiCl

41)

Consider the following attraction

(a) HCl------HCl (b) CH_4------CH_4

(c) NH_4^{+} ------ H_2O (d) He------He

The correct order of strength of given attraction is

(1) c > b > a > d

(2) c > a > b > d

(3) c > d > a > b

(4) c > b > d > a

42) In which of the following pairs, both species shows conductivity ?

(1) $AlF_{3(molten)}$, $AlCl_{3(molten)}$

(2) $AlCl_{3(aq.)}$, $Na_{(s)}$

(3) $NaCl_{(s)}$, $HCl_{(aq.)}$

(4) $HCl_{(molten)}$, $AlF_{3(aq.)}$

43) ***Assertion :-*** The single N–N bond is weaker than single P–P bond.
Reason :- Interelectronic repulsion between the two smaller N–N atoms weakens the N–N bond.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

44) Which of the following order of bond angle would be correct?

(1) $NO_2 > NO_2^+ > NO_2^-$

(2) $NO_2^- > NO_2^+ > NO_2$

(3) $NO_2^+ > NO_2 > NO_2^-$

(4) $NO_2^+ > NO_2^- > NO_2$

45) If s-p mixing is not allowed then species containing only σ bond :-

(1) B_2

(2) C_2

(3) N_2

(4) O_2

BIOLOGY

1) In animals steroidal hormones are synthesised by which organelle?

(1) Golgi bodies

(2) Nucleus

(3) Vacuole

(4) Endoplasmic reticulum

2) Match the column I and column II.

	Column-I		**Column-II**
(a)	Active transport	(i)	Movement of water by diffusion
(b)	Passive transport	(ii)	Na^+/K^+ pump
(c)	Osmosis	(iii)	Movement of neutral solutes

(1) (a)-(i), (b)-(ii), (c)-(iii)

(2) (a)-(iii), (b)-(ii), (c)-(i)

(3) (a)-(ii), (b)-(iii), (c)-(i)

(4) (a)-(i), (b)-(iii), (c)-(ii)

3) Select the correct statement regarding plant vacuoles.

(1) The vacuole is membrane less space found in the cytoplasm.

(2) Vacuole store water sap and excretory product.

(3) In plant cell vacuole is absent.

(4) In plant cell, tonoplast facilitates the transport of ion along the concentration gradients only.

4) Which of the following cell organelle is found in both prokaryotic and eukaryotic cell ?

(1) Endoplasmic reticulum
(2) Golgi complex
(3) Lysosome
(4) Ribosome

5) Depending on the _____ membrane proteins can be classified as integral and peripheral.

(1) Size
(2) Ease of extraction
(3) Position
(4) Shape

6)

Plasmodesmata helps in:

(1) Movement of water only
(2) Cytoplasmic connection between cells
(3) Nuclear division
(4) Cell wall formation

7) Result of compartmentalization in cytoplasm of Eukaryotic cell is :-

(1) Cell organelles
(2) DNA
(3) Cell wall
(4) Cell membrane

8) The centrosome is important for :

(1) DNA replication
(2) Microtubule organization during cell division
(3) Protein synthesis
(4) Lipid metabolism

9) What is the arrangement of microtubules in a typical eukaryotic cilium or flagellum :

(1) 9 + 0 pattern
(2) 9 + 2 pattern
(3) 9 + 1 pattern
(4) 10 + 2 pattern

10) Which of the following is false for nucleolous / Nucleoli ?

(1) not membrane bounded structure
(2) site for active r-RNA synthesis

(3) Nucleoli are spherical structures present in nucleoplasm
(4) Larger nucleoli are present in cells actively carrying out lipid synthesis

11) Recombinase enzyme activity is observed in :

(1) Zygotene
(2) Diplotene
(3) Pachytene
(4) Anaphase II

12) Formation and dissolution of synaptonemal complex occurs in respectively :

(1) Zygotene and Diplotene
(2) Pachytene and Diplotene
(3) Zygotene and Pachytene
(4) Zygotene and Diakinesis

13) Which of the following events is continuously occuring in interphase :-

(1) Histone protein formation
(2) Growth of cells
(3) DNA replication
(4) Centriole duplication

14) Centromere of each chromosome is toward the pole and arms of chromosome trailing behind it. This statement is related with :

(1) Prophase
(2) Metaphase
(3) Anaphase
(4) Telophaste

15) Which one of the following statements is not correct regarding Amino acids.

(1) Amino acids are substituted methanes.
(2) The R group in proteinaceous amino acids could be a methyl group (eg : Glycine).
(3) In solutions of different pH, the structure of amino acids changes.
(4) $-NH_2$ & –COOH groups are of ionisable nature.

16) Long protein chain folded upon itself like hollow woolen ball structure known as :-

(1) Primary structure
(2) Secondary structure
(3) Tertiary structure
(4) None of these

17) GLUT-4 :-

(1) Produces rubber, gums and cellulose after polymerisation
(2) Enables glucose transport into cells
(3) Is the most abundant protein in whole biosphere
(4) Is a complex heteropolymer of fructose.

18) Match the correct amino acid according to it's nature :

	Acidic	**Alkaline**	**Neutral**
(1)	Glutamic acid	Arginine	Histidine
(2)	Palmitic acid	Histidine	Valine
(3)	Aspartic acid	Alanine	Lysine
(4)	Glutamic acid	Lysine	Valine

(1) 1
(2) 2
(3) 3
(4) 4

19) According to enzyme substrate complex theory :-
Enzyme + Substrate $\rightleftharpoons$ A $\rightarrow$ B $\rightarrow$ E + P.
In the following reaction A and B is :-

(1) A $\rightarrow$ E.P., B $\rightarrow$ E.S.
(2) A $\rightarrow$ E.S., B $\rightarrow$ E.P.
(3) A $\rightarrow$ E.P., B $\rightarrow$ E.P.
(4) A $\rightarrow$ E.P., B $\rightarrow$ E.I.

20) An organic compound loosely bound to apoenzyme is known as :

(1) Metal ion
(2) Apoenzyme
(3) Haloenzyme
(4) Coenzyme

21) **Statement I** : Enzymes have been classified into different groups based on the type of reactions they catalyse.
Statement II : Hydrolases enzyme catalyses hydrolysis of ester, ether, peptide and glycosidic bonds.

(1) Both Statement I and Statement II are incorrect.
(2) Statement I is correct and Statement II is incorrect.
(3) Both Statement I and Statement II are correct.
(4) Statement I is incorrect and Statement II is correct.

22) Read the statement carefully and select the option which include correct statement -
(A) The activity of an enzyme can be affected by a change in the conditions which alter the tertiary structure of protein.
(B) Enzymes generally function in a narrow range of temperature and pH.
(C) Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys the enzyme activity.
(D) When the inhibitor closely resemble the substrate in its molecular structure and inhibit the activity of enzyme it is known as competitive inhibitor.
Option :

(1) Only A & B

(2) Only B & C

(3) Only B, C & D

(4) All are correct

23) What is the correct sequence for catalytic cycle of an enzyme action ?
(i) The enzyme releases the product.
(ii) The active site of enzyme is in close proximity of the substrate and breaks the chemical bonds of the substrate.
(iii) The binding of the substrate induces the enzyme to alter its shape.
(iv) The substrate binds to the active site of the enzyme.

(1) iv, iii, ii, i

(2) iii, ii, i, iv

(3) iv, ii, i, iii

(4) ii, i, iv, iii

24) **Statement – I :** Polysaccharides are long chains of sugars. They are threads containing different monosaccharides as building blocks.
Statement – II : The acid soluble pool represents only the cytoplasmic composition.
Read the above statements and select correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

25) Match column-I with column-II and select the correct option given below :

Column-I		**Column-II**	
(A)	Endothermic	(i)	Difference between energy of substrate from that of transition state
(B)	Exothermic	(ii)	Transition state

(C)	ES complex	(iii)	Energy level of product is lower than substrate.
(D)	Activation energy	(iv)	Energy level of product is higher than substrate

(1) (A)-iii, (B)-iv, (C)-ii, (D)-i

(2) (A)-iii, (B)-iv, (C)-i, (D)-ii

(3) (A)-iv, (B)-iii, (C)-ii, (D)-i

(4) (A)-iv, (B)-iii, (C)-i, (D)-ii

26) (A) Enzyme eventually bring down activation energy barrier making the transition of substrate to product easy.
(B) Enzyme substrate complex is highly reactive and this complex is short lived.
(C) Apoenzyme have two parts, protein part is called holoenzyme and non protein part is called co-factor.
(D) A general rule of thumb is that rate double or decrease by half for every 20°C change in either direction.
Select the correct option :

(1) A, B and C

(2) B, C and D

(3) Only A and B

(4) Only C and D

27) **Statement– I** :- Oxidoreductase and dehydrogenases both are same classes of enzyme because both are catalysing a transfer of a group other than hydrogen.
Statement– II :-Lyases and hydrolases both are different classes of enzyme, which catalyse hydrolysis of substrates and removal of group from substrate by mechanism other than hydrolysis respectively.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) Only **Statement I** is correct

(4) Only **Statement II** is correct

28) Match list I with list II

List-I		**List-II**	
A.	Alkaloids	I	Vinblastin
B.	Toxins	II	Morphine
C.	Drugs	III	Rubber
D.	Polymeric substances	IV	Cellulose
		V	Ricin

		VI	Codeine

Choose the correct Answer from options given below.

(1) A-(II,VI) ; B-(V) ; C-(I) ; D-(III,IV)

(2) A-(II,V) ; B-(I) ; C-(III) ; D-(IV, VI)

(3) A-(III,IV) ; B-(I,V) ; C-(II) ; D-(VI)

(4) A-(II,VI) ; B-(I) ; C-(V) ; D-(II,IV)

29) Correctly match column-I, II and III.

	Column-I		**Column-II**		**Column-III**
(A)	Golgi apparatus	(P)	Chromatin	(i)	Protein synthesis
(B)	Ribosome	(Q)	Cristae	(ii)	Packaging and secretion
(C)	Nucleus	(R)	Cisternae	(iii)	Controller of cell
(D)	Mitochondria	(S)	RNA and proteins	(iv)	Aerobic respiration

(1) (A)-(P)-(i), (B)-(Q)-(ii), (C)-(R)-(iii), (D)-(S)-(iv)

(2) (A)-(R)-(ii), (B)-(P)-(i), (C)-(S)-(iii), (D)-(Q)-(iv)

(3) (A)-(R)-(ii), (B)-(S)-(i), (C)-(P)-(iii), (D)-(Q)-(iv)

(4) (A)-(S)-(ii), (B)-(R)-(i), (C)-(P)-(iii), (D)-(Q)-(iv)

30) How many given below structures are commonly found in both flagella and centrioles ?
Radial spokes, Hub, Plasma membrane, Peripheral microtubules, central microtubules

(1) Four

(2) Three

(3) Two

(4) One

31) Given below are two statements:
Statement-I :- The prokaryotic cells are represented by bacteria, blue-green algae, Mycoplasma and PPLO. They are generally smaller and multiply more rapidly than the eukaryotic cells.
Statement-II :- Prokaryotes have a cell wall surrounding the cell membrane except in Mycoplasma. The semifluid matrix filling the cell is the cytoplasm. There is no well-defined nucleus. The genetic material is basically naked, not enveloped by a nuclear membrane.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

32) **Assertion :-** The complete disintegration of the nuclear envelope marks the start of the third phase of mitosis, hence the chromosome are spread through the cytoplasm of the cell.
Reason :- Metaphase is characterized by all chromosome coming to lie at the equator with one chromatid of each chromosome connected by its kinetochore to spindle fibres from one pole and its sister chromatid connected by its kinetochore to spindle fibres from the opposite pole.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

33) In a diploid cell before S-phase quantity of DNA is 40 pico gram (pg) and chromosome number is 40 then what will be the amount of DNA and chromosome number in prophase-I, anaphase-I, after meiosis-I and meiosis-II ?

	Prophase-I	**Anaphase-I**	**After meiosis-I**	**After meiosis-II**
(1)	40 pg, 40	40 pg, 40	20 pg, 40	20 pg, 20
(2)	80 pg, 40	40 pg, 40	20 pg, 20	10 pg, 10
(3)	40 pg, 40	20 pg, 40	20 pg, 20	10 pg, 10
(4)	80 pg, 40	80 pg, 40	40 pg, 20	20 pg, 20

(1) 1

(2) 2

(3) 3

(4) 4

34) Which one of the following statement regarding S-phase is correct :

(1) Cell is metabolically active and continuously grows but does not replicate its DNA

(2) The amount of DNA per cell remains constant

(3) There is no increase in the chromosome number

(4) The amount of DNA and chromosome number both becomes double

35) Match the list-A with list-B and select correct answer from given options:

List-A		**List-B**	
(A)	Prophase	(i)	Chromosome lost their identify
(B)	Metaphase	(ii)	Golgi body, nucleolus and nuclear membrane disappear
(C)	Anaphase	(iii)	Condensation of chromosome is complete

(D)	Telophase	(iv)	Chromatids move to opposite poles.

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-II, C-IV, D-I

(4) A-II, B-III, C-IV, D-I

36) Match List-I with List-II

	List-I (Events)		**List-II (Phases of Meiosis)**
(A)	The bivalent chromosomes align on equatorial plant	(I)	Anaphase-I
(B)	The homologous chromosome separate while sister chromatids remain associated at their centromere	(II)	Pachytene
(C)	In many case, chromosome do undergo some dispersion, they donot reach extremely extended state of interphase nucleus	(III)	Metaphase-I
(D)	Bivalent or Tetrad become clearly visible	(IV)	Telophase-I

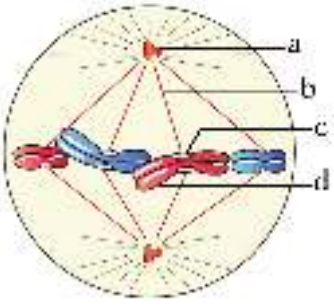
Choose the correct answer from the options given below:-

(1) A-I, B-II, C-III, D-IV

(2) A-III, B-I, C-II, D-IV

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-III, C-II, D-I



37) Study the following diagram carefully and identify the correct match:

(1) a : Centromere

(2) b : Spindle fibre

(3) c : Centriole

(4) d : Tetrad

38) Some events of mitotic cell division are given below in unordered manner -
(i) Condensation of chromatin is completed
(ii) Chromosome lose their identity as discrete elements
(iii) Sister chromatid separate
(iv) Initiation of the assembly of mitotic spindle between pair of centrioles
The correct order for steps of cell division is :-

(1) i → iv → iii → ii

(2) iv → i → iii → ii

(3) ii → i → iv → iii

(4) iv → iii → i → ii

39) **Assertion :-** Ribosomes are the granular structures composed of ribonucleic acid (RNA) and proteins and are not surrounded by any membrane.
Reason :- Sedimentation coefficient is indirectly a measure of density and size. Both 70S and 80S ribosomes are composed of two subunits.

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

40) **Assertion (A) :** Cell is the fundamental structural and functional unit of all living organism.
Reason (R) : Unicellular organism are capable of (i) independent existence and (ii) performing the essential function of life.
Anything less than a complete cell of does not ensure independent living.

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**.

(3) **A** is true but **R** is false.

(4) **A** is false but **R** is true.

41) Identify the correct statement –
(a) E.R. principally performs the function of packaging material.
(b) Lysosome is bound by a single membrane called tonoplast.
(c) Amyloplast store carbohydrate.
(d) The inner compartment of mitochondrial is filled with matrix.

(1) (a), (b), (c) & (d)

(2) (a), (b) & (d)

(3) (c) & (d)

(4) (a) & (d)

42) Select the correct statement ?

(1) According to fluid mosaic model, the quasi fluid nature of protein enables the lateral movement of lipids within the overall bilayer

(2) One of the most important functions of the cell wall is the transport of the molecules across it

(3) In cell membrane of different organisms, the ratio of protein and lipid varies considreably in different cell types

(4) The cell membrane and middle lamella in animal cell may be traversed by plasmodesmata which connect the cytoplasm of neighbouring cells.

43) Read the statement carefully
(A) Chlorophyll pigments are present in thylakoids.
(B) Ribosomes of chloroplast are larger than cytoplasmic ribosomes.
(C) Number of chloroplast in *chlamydomonas* varies between 20 and 40.
(D) Centrioles acts as site of active RNA synthesis
How many of the above statement are correct.

(1) One

(2) Two

(3) Three

(4) Four

44) Match column I with column II and select correct option :

	Column-I		**Column-II**
(A)	Mesosome	(i)	Extension of plasma membrane into the cells
(B)	Polysome	(ii)	Translation of mRNA into proteins
(C)	Centriole	(iii)	Nonmembranous cell organelle
(D)	Nuclear envelop presence	(iv)	Eukaryotic cell

(1) A-i, B-ii, C-iii, D-iv

(2) A-ii, B-i, C-iii, D-iv

(3) A-i, B-ii, C-iv, D-iii

(4) A-ii, B-i, C-iv, D-iii

45) Read the following terminology of column-I & column-II and identify their correct match :

	Column-I		**Column-II**
(A)	Chromatin	(i)	Disc shaped structure on centromere
(B)	Histones	(ii)	Nucleolus and chromatin

(C)	Kinetochore	(iii)	Nucleoprotein fibres
(D)	Nucleoplasm	(iv)	Basic protein
		(v)	Satellite

(1) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)

(2) (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)

(3) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

(4) (A)-(i), (B)-(ii), (C)-(iv), (D)-(iii)

46) Mitral valve in mammals guards the opening between :-

(1) Right atrium and Right ventricle

(2) Left atrium and Left ventricle

(3) Right atrium and Left ventricle

(4) Left atrium and Right ventricle

47) Mark the odd one w.r.t white blood cells :-

(1) Monocytes

(2) Neutrophils

(3) Erythrocytes

(4) Lymphocytes

48)

Another mass of nodal tissue called
atrio-ventricular node (AVN) is present in :-

(1) Right upper corner of right atrium

(2) Left upper corner of right atrium

(3) Right lower corner of left atrium

(4) Left lower corner of right atrium

49) A portal vein :-

(1) starts from an organ and ends in heart

(2) divides in heart and restarts by the union of its capillaries

(3) collects blood from the gut and pours it into the inferior vena cava

(4) drains blood from an organ and pours it into another organ

50) First heart sound is :

(1) Lubb sound at the end of systole

(2) Lubb sound at the beginning of ventricular systole

(3) Dup sound at the end of systole

(4) Dup sound at the beginning of ventricular systole

51) A special system of blood vessels present for circulation of blood to and from the cardiac musculature is called as :-

(1) Hepatic portal system
(2) Hypophyseal portal system
(3) Coronary system
(4) Cutaneous circulation

52) Time interval between the closure of semilunar valve and closure of AV valve is

(1) 0.3 sec
(2) 0.5 sec
(3) 0.1 sec
(4) 0.7 sec

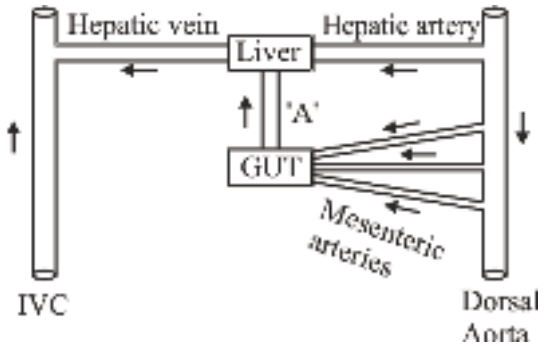
53) Plasma protein primarily involved in defense mechanism of the body :-

(1) Albumins
(2) Globulins
(3) Fibrinogen
(4) Prothrombin

54) **Assertion :** If repeated checks of blood pressure of an individual is 140/90 or higher, it shows hypertension.
Reason : Hypotension is the term for blood pressure that is higher than normal.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

55)



Identify 'A' in above figure ?

(1) Hepatic duct
(2) Hepatic portal vein
(3) Branch of bile duct
(4) Vassa vassorum

56) Cardiac activity could be moderated by the autonomous neural system. Find out the correct answer:-

(1) The parasympathetic system stimulates heart rate and stroke volume

(2) The sympathetic system stimulates heart rate and stroke volume

(3) The parasympathetic system decreases the heart rate but increases the stroke volume

(4) The sympathetic system decreases the heart rate but increase stroke volume

57) Which of the following is mismatched ?

(1) **Lubb :** Ist heart sound associated with closure of tricuspid and bicuspid valves.

(2) **Cardiac output :** Stroke volume multiplied by heart rate.

(3) **Dup :** Second heart sound, due to opening of semilunar valves.

(4) **Duration of cardiac cycle :** 0.8 second.

58) **Statement-I :** In *atherosclerosis* only deposition of fat and fibrous tissues occurs, which make the wall of arteries hard.
Statement-II : Only reason of heart failure is congestion of liver.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

59)

Match the Column-I with Column-II :-

Column-I		**Column-II**	
(A)	Inter ventricular septum	(i)	Thin, fibrous
(B)	Atrio ventricular septum	(ii)	Thick, fibrous
(C)	Inter atrial septum	(iii)	Thin, muscular
		(iv)	Thick, muscular

(1) A-(iv), B-(i), C-(iii)

(2) A-(ii), B-(iv), C-(i)

(3) A-(iv), B-(ii), C-(iii)

(4) A-(ii), B-(iv), C-(iii)

60) Match the Column-I with Column-II.

	Column-I		**Column-II**
(A)	Fish	(i)	3-chambered heart
(B)	Amphibia	(ii)	Incomplete double circulation

(C)	Birds	(iii)	4-chambered heart
		(iv)	Single circulation
		(v)	2-chambered heart
		(vi)	Double circulation

(1) A→(i), (ii) B→(iii), (vi) C→(iv), (v)

(2) A→ (i), (iv) B→(v), (ii) C→(iii), (vi)

(3) A→(v), (iv) B→(i), (ii) C→(iii), (vi)

(4) A→(iii), (ii) B→(i), (iv) C→(v), (vi)

61) Consider the following statements (a–d) and select the option which includes all the correct ones only:-
(a) At the time of ventricular systole atrioventricular valve are closed and semilunar valve are opened.
(b) At the time of ventricular diastole atrioventricular valve are opened and semilunar valve are closed.
(c) At the time of joint diastole atrioventricular valve are opened and semilunar valve are closed.
(d) At the time of atrial systole atrioventricular valve are closed and semilunar valve are opened.
Options :-

(1) Statements a, b and c

(2) Statements a and d

(3) Statements c and d

(4) Statements b, c and d

62) (a) SAN is responsible for initiating and maintaining the rhythmic contractile activity of heart
(b) Body has ability to alter the stroke volume as well as heart rate and there by the cardiac output
(c) Cardiac output of an athlete will be much lower than that of an ordinary man
(d) Normal activities of the heart are regulated intrinsically
Mark the correct sequence of true and false

(1) T, F, F, T

(2) F, T, T, F

(3) T, T, F, T

(4) T, F, T, F

63) What will be the P_{O_2} and P_{CO_2} in the atmospheric air compared to those in the alveolar air ?

(1) P_{O_2} lesser, P_{CO_2} higher

(2) P_{O_2} higher, P_{CO_2} lesser

(3) P_{O_2} higher, P_{CO_2} higher

(4) P_{O_2} lesser, P_{CO_2} lesser

64) Shape of diaphragm is -

(1) Dome - Shaped
(2) U - Shaped
(3) C - Shaped
(4) Ring - Shaped

65) Which of the following epithelium is present in diffusion membrane ?

(1) Simple cuboidal epithelium
(2) Ciliated epithelium
(3) Compound epithelium
(4) Simple squamous epithelium

66) About____percent of O_2 is transported by R.B.C. & ____ % O_2 is carried by plasma :

(1) 93, 7
(2) 97, 3
(3) 70, 25
(4) 40, 60

67) A chemosensitive area adjacent to the rhythm centre is highly sensitive to _____ and hydrogen ions.

(1) Oxygen
(2) Carbondioxide
(3) Nitrogen
(4) CO

68) Where is the respiratory rhythm centre located in the brain ?

(1) Cerebellum
(2) Medulla oblongata
(3) Pons
(4) Hypothalamus

69) The role of ____ in regulation of respiration is quite insignificant.

(1) CO_2
(2) H^+
(3) O_2
(4) Both (1) & (2)

70) **Assertion :** In certain industries, so much dust is produced that the defence mechanism of the body can not fully cope with the situation.
Reason : Long exposure of dust can give rise to inflammations leading to emphysema and thus causing serious lung damage.

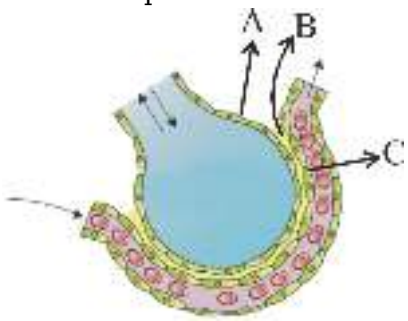
(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(2) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

71) Given below is a diagram that shows section of alveoli along with alveolar capillary. In this diagram A, B and C compose the diffusion-membrane for gaseous exchange. What are the A, B and C respectively?



	A	B	C
(1)	Capillary endothelium	Alveolar epithelium	Basement substance
(2)	Alveolar epithelium	Capillary endothelium	Basement substance
(3)	Basement substance	Alveolar epithelium	Capillary endothelium
(4)	Alveolar epithelium	Basement substance	Capillary endothelium

(1) 1

(2) 2

(3) 3

(4) 4

72) Which of the following statement is correct?

(1) Sponges, Coelenterates, Flat worms etc exchange O_2 with CO_2 by simple diffusion over their entire body surface.

(2) Earthworm has tracheal system to transport atmospheric air within the body.

(3) Gills are used by insects for respiration.

(4) Molluscs respire by lungs.

73) The chloride content of RBC will be higher in :-

(1) Systemic arteries and pulmonary veins

(2) Systemic veins and pulmonary arteries

(3) Systemic arteries and pulmonary arteries

(4) Systemic veins and pulmonary veins

74) Match the following and mark the **correct** options :

	Animal		**Respiratory Organ**
A.	Earthworm	i.	Moist cuticle
B.	Insects	ii.	Gills
C.	Fishes	iii.	Lungs
D.	Birds/Reptiles	iv.	Tracheal tube

(1) A-ii, B-i, C-iv, D-iii

(2) A-i, B-iv, C-ii, D-iii

(3) A-i, B-iii, C-ii, D-iv

(4) A-i, B-ii, C-iv, D-iii

75) Given below are two statements :
Statement-I :- Mechanisms of breathing vary among different group of animals depending mainly on their habitats and level of organisation.
Statement-II :- Amphibians like frogs can respire through their moist cuticle.
In the light of above statements, choose the correct answer from the option given below :-

(1) Statement I is true but statement II is false.

(2) Statement I is false but statement II is true.

(3) Both statement I and statement II are true.

(4) Both statement I and statement II are false.

76) Read column-I, II, & III and identify their correct match -

	Column-I		**Column-II**		**Column-III**
A	Larynx	P	Frog	i.	Incomplete cartilagineous ring
B	Hibernation	Q	Catabolic process	ii.	Sound production
C	Cellular respiration	R	Cartilaginous box	iii.	Cutaneous respiration
D.	Trachea	S	Wind pipe	iv	Liberation of energy

(1) A–R–ii, B–P–iii, C–Q–iv, D–S–i

(2) A–R–iv, B–R–ii, C–P–i, D–R–iii

(3) A–P–iii, B–P–iv, C–Q–ii, D–S–i

(4) A–R–ii, B–P–iii, C–S–iv, D–Q–i

77) Which is the correct sequence of the air passage in man?

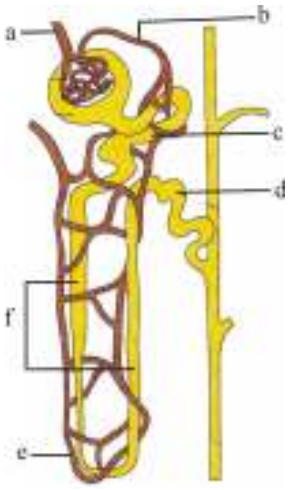
(1) Nasal cavity → pharynx → trachea → larynx → bronchi → bronchioles → alveoli

(2) Nasal cavity → pharynx → larynx → trachea → bronchi → bronchioles → alveoli

(3) Nasal cavity → larynx → pharynx → trachea → bronchi → bronchioles → alveoli

(4) Nasal cavity → larynx → bronchi → pharynx → trachea → bronchioles → alveoli

78) Recognise the figure and find out the correct matching :-



(1) c-PCT, d-DCT, a-afferent arteriole, b-efferent arteriole, f–henle's loop, e-vasa recta

(2) d-PCT, c-DCT, b-afferent arteriole, a-efferent arteriole, e-henle's loop, f-vasa recta

(3) c-PCT, d-DCT, b-afferent arteriole, a-efferent arteriole, f–henle's loop, e-vasa recta

(4) d-PCT, c-DCT, a-afferent arteriole, b-efferent arteriole, e-henle's loop, f-vasa recta

79) How many of the following statements are true?

Statement 1 : Blood is pumped into a dialyzing unit, where it passes through a coiled cellophane tube surrounded by a fluid (dialyzing fluid) with the same composition as plasma, except for nitrogenous wastes.

Statement 2 : The porous cellophane membrane allows the removal of urea and other substances from the blood in the dialyzing unit.

Statement 3 : The composition of the dialyzing fluid is similar to that of plasma, excluding nitrogenous wastes.

Statement 4 : An anti-coagulant like heparin is added to the blood before it is returned to the body during the process of hemodialysis.

(1) 1

(2) 2

(3) 4

(4) 3

80) **Assertion :** Diabetes insipidus is marked by excessive urination & too much thirst for water.

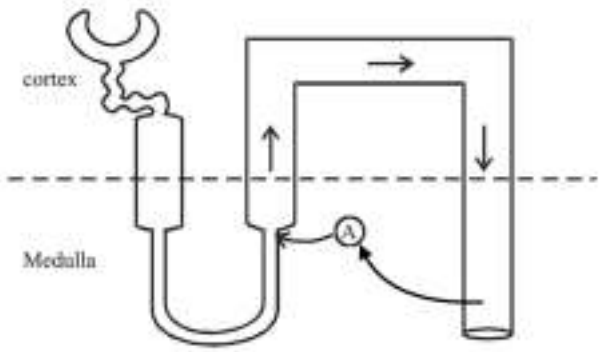
Reason : ADH is secreted by posterior lobe of pituitary gland.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

81) 

In the given figure substance A name and its work is:-

(1) NH_3 - Increases osmolarity in medullary part

(2) NaCl - Increases osmolarity in medullary part

(3) Urea - Increases osmolarity in medullary part

(4) H_2O - Increases osmolarity in medullary part

82) Which of the following statements is correct ?

(1) Human excretes 1 to 1.5 lit/day urine.

(2) The colour of urine is pale yellow & slightly alkaline in nature.

(3) Minimum Reabsorption occurs in PCT.

(4) Glucose and amino acid are reabsorbed by passive transport.

83) Match the following parts of a nephron with their function :-

(a)	Descending limb of Henle's loop	(i)	Reabsorption of salts only
(b)	Proximal Convoluted tubule	(ii)	Reabsorption of water only
(c)	Ascending limb of Henle's loop	(iii)	Conditional reabsorption of sodium ion & water
(d)	Distal convoluted tubule	(iv)	Reabsorption of ion, water and organic nutrients.

Select the correct option from the following :

(1) (a)-(i), (b)-(iii), (c)-(ii), d-(iv)

(2) (a)-(ii), (b)-(iv), (c)-(i), d-(iii)

(3) (a)-(i), (b)-(iv), (c)-(ii), d-(iii)

(4) (a)-(iv), (b)-(i), (c)-(iii), d-(ii)

84)

Nitrogenous waste products are eliminated mainly as -

(1) Urea in tadpole & ammonia in adult frog
(2) Ammonia in tadpole and urea in adult frog
(3) Urea in both tadpole & adult frog.
(4) Urea in tadpole and uric acid in adult frog.

85) Brush border is characteristic of –

(1) Neck of nephron
(2) Collecting tube & Glomerulus
(3) Proximal convoluted tubule
(4) Bowman capsule

86) GFR is increased by :

(1) ADH
(2) GH
(3) Renin
(4) Both (1) and (3)

87) In a healthy person the glomerular filtration rate GFR per day will be :-

(1) 125 ml
(2) 650 ml
(3) 180 litre
(4) 170 litre

88) The conditional reabsorption of Na^+ and water occurs in which part of nephron ?

(1) PCT
(2) Loop of Henle
(3) Bowman's capsule
(4) DCT

89) Which of the following structures are found in flatworms :

(1) Antennal glands
(2) Malpighian tubules
(3) Nephridia
(4) Flame cells

90) The fluid entering the proximal tubule of nephron has the osmolarity

(1) Higher than plasma
(2) Same as plasma
(3) Lesser than plasma
(4) Same as innermost interstitial medulla

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	2	1	2	2	3	1	4	1	1	3	1	2	2	4	3	2	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	1	2	4	3	3	4	1	3	3	4	2	2	1	4	3	1	3	2
Q.	41	42	43	44	45															
A.	1	3	4	4	1															

CHEMISTRY

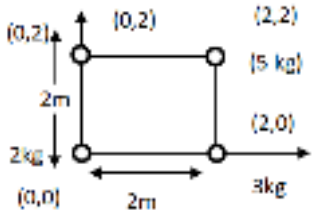
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	1	1	3	3	2	4	1	1	1	2	3	4	3	3	4	3	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	4	3	4	4	2	4	1	1	3	4	3	2	2	3	1	1	2	2
Q.	86	87	88	89	90															
A.	2	2	4	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	2	4	2	2	1	2	2	4	3	1	2	3	2	3	2	4	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	1	1	3	3	2	1	3	3	3	3	4	3	4	3	2	2	1	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	1	1	1	2	3	4	4	2	3	2	2	3	2	2	3	4	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	3	2	1	4	2	2	2	3	3	4	1	2	2	3	1	2	1	4	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	2	2	3	4	3	4	4	2										

SOLUTIONS

PHYSICS



1) $x_{com} = \dfrac{2(0)+3(2)+5(2)+8(0)}{2+3+5+8}$

$= \dfrac{16}{18}$

$x_{com} = \dfrac{8}{9}$

$y_{com} = \dfrac{2(0)+3(0)+5(2)+8(2)}{2+3+5+8}$

$= \dfrac{26}{18}$

so $com(\dfrac{8}{9}, \dfrac{13}{9})$

2) By momentum conservation

$M100\cos 37^\circ = \dfrac{M}{4}(0) + \dfrac{3M}{4}\vec{v}\ \vec{v} = \dfrac{320}{3}$ m/s

3) $a_{cm} = \dfrac{F_{net}}{M_{total}} = \dfrac{30}{10+20} = 1 m/s^2$

$S = u_{cm}t + \dfrac{1}{2}a_{cm}t^2$

$= 0 + \dfrac{1}{2}(1)(2)^2 = 2m$

4)

Impulse = Area under F-t graph

$e = \dfrac{\text{Impulse of restitution}}{\text{Impulse of deformation}} = \dfrac{\frac{1}{2} \times 0.36 \times 5000}{\frac{1}{2} \times 0.64 \times 5000}$

$e = \dfrac{9}{16} = 0.5625$

5) Since collision is elastic, 3 Kg blocks will exchange their velocities.

$v_{cm} = \dfrac{m_1v_1 + m_2v_2}{m_1 + m_2} = \dfrac{3 \times 2 + 6 \times 0}{3+6} = \dfrac{2}{3} m/s$

Using energy conservation

$\dfrac{1}{2} \times 3 \times 2^2 = \dfrac{1}{2} \times Kx^2 + \dfrac{1}{2}(3+6)v_{cm}^2$

x = 20 cm

6) This question is based on for frictionless banking :

$v = \sqrt{Rg\tan\theta},\ \tan\theta = \frac{h}{b}$

$\tan\theta = \frac{2}{20} = 0.1$

$v = \sqrt{15 \times 9.8 \times 0.1} = \sqrt{14.7}$ m/s

7) For uniform circular motion:

$\omega = \frac{2\pi}{T},\ a_C = \omega^2 r$

r = 10cm T=π s

$\omega = \frac{2\pi}{T} = 2$ rad / s

$a_c = \omega^2 r = 4 \times 10 = 40 cm / s^2$

8) $V_\perp = \omega r$

$V\sin\theta = \omega\left(\frac{3}{\sin\theta}\right)$

$V = (\omega)\frac{3}{\sin^2\theta}$

$V = \frac{(0.1)(3)}{\left(\frac{1}{2}\right)} = 0.6$ m/s

9)

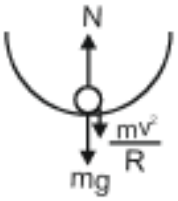
$T - mg = \frac{mv^2}{\ell}$ (At mean position)

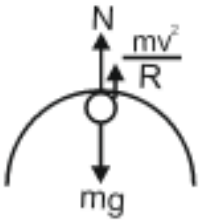
$\Rightarrow 3mg - mg = \frac{mv^2}{\ell}$

$\Rightarrow v = \sqrt{2g\ell}$

With this speed, the pendulum becomes horizontal at extreme position.

$\therefore \theta = 90°$





10)

$N_A = mg + \frac{mv^2}{R_A}$

$N_B = mg - \frac{mv^2}{R_B}$

$$N_C = mg + \frac{mv^2}{R_C}$$

From the curve, $R_A < R_C \Rightarrow N_A > N_C \Rightarrow N_A$ is maximum

11) $J = mvr$

$$= m\left(\frac{2\pi r}{T}\right) r$$

$$= \frac{2m(\pi r^2)}{T}$$

$$J = \frac{2mA}{T}$$

12) K.E. $= \frac{L^2}{2mr^2}$

P.E. $= -\frac{L^2}{mr^2}$

T.E. $= -\frac{L^2}{2mr^2}$

13)

There is no effect of rotation of earth on its axis on poles.

$$g_h = g\left(1 - \frac{2h}{R}\right)$$

14) $\frac{W}{m_0} = \vec{I}.\vec{dr}$

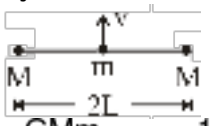
$\vec{dr} = 5\hat{i} + 4\hat{j}$

$\frac{W}{2} = 10(\hat{i} + \hat{j}).(5\hat{i} + 4\hat{j})$

$W = 2(50 + 40)\ J = 180\ J$

15)

By C.O.M.E.



$$-\frac{GMm}{L} \times 2 + \frac{1}{2}mv^2 = 0$$

$$\frac{1}{2}mv^2 = \frac{2GMm}{L} \Rightarrow v = 2\sqrt{\frac{GM}{L}}$$

16) $U_{surface} = -mgR_e$

$$U_{h=R_e} = \frac{-GMm}{R_e + R_e} = \frac{-GMm}{2R_e}$$

$$U_{h=R_e} = \frac{-mgR_e}{2}$$

17) Orbital radius : $r_1 = R + R = 2R$

$r_2 = R + 7R = 8R$

$$KE_1 = \frac{GMm}{2r_1} = \frac{GMm}{2(2R)}$$

$$KE_2 = \frac{GMm}{2r_2} = \frac{GMm}{2(8R)}$$

$$\frac{KE_1}{KE_2} = 4$$

18) $$W = m \times \frac{GM}{r^2}$$

$$W' = m \times \frac{GM}{(2r)^2}$$

$$W' = \frac{W}{4}$$

19) $$\Delta U = W = \frac{-GmM}{5x} + \frac{GmM}{x}$$

$$= \frac{4GmM}{5x}$$

20) $$h = \frac{2T\cos\theta}{\rho rg}$$

A → Soap decreases the surface tension so h decreases

B → In free fall lift

$g_{eff} = 0$

h = Full length

C → $g_{eff} = (g + a) \Rightarrow h \downarrow$

D → $g_{eff} = (g - a) \Rightarrow h \uparrow$

21)

$$P_A = p_0 + \rho g h_1 = (10^5) + (1000)(10)(10)$$

$$P_A = 2 \times 10^5 \text{ Pa}$$

$$P_B = P_A + (2\rho)gh_2 = (2 \times 10^5) + (2 \times 10^3)(10)(20)$$

$$P_B = 6 \times 10^5 \text{ Pa}$$

Ratio : $$\frac{P_A}{P_B} = \frac{1}{3}$$

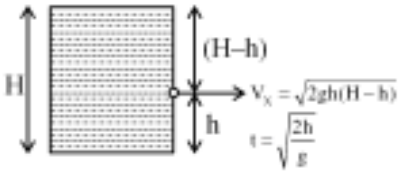
22)

$P_1 = P_2$

$\rho_{oil}\, gh + \rho_{Hg}\, g(20 - h) = \rho_c g\, (20)$

$\Rightarrow 0.9 \times h + 13.6\,(20 - h) = 1.6 \times 20$

⇒ 20(13.6 – 1.6) = h (13.6 – 0.9)
⇒ h = 18.9 cm

23) 

Range (R) = $V_x t = \sqrt{2g(H-h)}\sqrt{\frac{2h}{g}}$
When hole at a height h_1 from bottom

$$R_1 = \sqrt{2g(H-h_1)}\sqrt{\frac{2h_1}{g}}$$

When hole at a height h_2 from bottom

$$R_2 = \sqrt{2g(H-h_2)}\sqrt{\frac{2h_2}{g}}$$

according to question, ranges are equal
$R_1 = R_2$

$$\sqrt{2g(H-h_1)}\sqrt{\frac{2h_1}{g}} = \sqrt{2g(H-h_1)}\sqrt{\frac{2h_2}{g}}$$

⇒ $(H - h_2)h_2 = (H - h_1)h_1$
$\Rightarrow h_1^2 - h_2^2 = H(h_1 - h_2)$
⇒ $(h_1 + h_2)(h_1 - h_2) = H(h_1 - h_2)$
⇒ $H = h_1 + h_2$

24)

$A_1v_1 = A_2v_2$
$v_1 = v_2 \Rightarrow$ (A = constant)

25)

$A_1v_1 = A_2v_2$

$$v_2 = \frac{A_1v_1}{A_2} = 20$$

26) $Y = 10^4 N/m^2$, $A=2 \times 10^{-4} m^2$,
$F = 2 \times 10^5$dyne = 2N

$$\Delta\ell = \frac{FL}{AY} = \frac{2 \times L}{2 \times 10^{-4} \times 10^4} = L$$

 Final length = initial length + increment = 2L

27) $P = B\frac{\Delta V}{V}$

$$\frac{\Delta V}{V} = \frac{P}{B} = \frac{1 \times 10^5}{1.25 \times 10^{11}} = 8 \times 10^{-7}$$

28) $\Delta \ell = \dfrac{F\ell}{Ay}$

Slope = $\dfrac{\ell}{Ay} \Rightarrow$ Slope = $\dfrac{\ell A}{A^2 y}$

$= \dfrac{\text{Vol.}}{A^2 y}$

Slope $\propto \dfrac{1}{r^4}$

29) Work done in stretching the wire through 0.61 mm under the load of 3 kg wt.

$W_1 = \dfrac{1}{2}$ stretching force × extension

$= \dfrac{1}{2} \times 3 \times 9.8 \times 0.61 \times 10^{-3}$

$= 8.967 \times 10^{-3}$ J

Work done in stretching the wire through 1.02 mm under the load of 5 kg wt.

$W_2 = \dfrac{1}{2} \times 5 \times 9.8 \times 1.02 \times 10^{-3}$

$= 24.99 \times 10^{-3}$ J

Hence the work done in stretching the wire from 0.61 mm to 1.02 mm.

$\Delta W = W_2 - W_1 = (24.99 - 8.961) \times 10^{-3}$

$\simeq 16 \times 10^{-3}$ J

30) $\beta = \dfrac{-\Delta P}{(\Delta V/V)}$

For incompressible liquid $\dfrac{\Delta V}{V} = 0$

$\boxed{B \to \infty}$

31) $V_T = \dfrac{2}{9} \dfrac{r^2(\rho - \sigma)}{\eta} g$

$V_T \propto (\rho - \sigma) . r^2$

So,

$$\frac{V_1}{V_2} = \left[\frac{\rho - \sigma_1}{\rho - \sigma_2}\right] \cdot \left[\frac{r_1}{r_2}\right]^2 = \frac{13.6 - 7.6}{13.6 - 11.6} \times \left(\frac{1}{2}\right)^2$$

$$\frac{V_1}{V_2} = \frac{3}{4} = 0.75$$

32)

$Q = \dfrac{\pi P r^4}{8\eta L}$ ....(i)

$2Q = \frac{\pi P'(2r)^4}{8\eta(2L)}$ ....(ii)

(ii) ÷ (i) $\frac{2}{1} = \frac{(P')(8)}{(P)(1)}$

$P' = \frac{P}{4}$

33) $h = \frac{2T\cos\theta}{rdg} \therefore h \propto \frac{1}{r}$ So the graph between h and r will be rectangular hyperbola.

34)

$P_{ex} = \frac{4T}{r}$ ...(1)

$\Rightarrow \frac{4T}{r} = \rho gh$

$T = \frac{\rho ghr}{4} = \frac{0.8 \times 1000 \times 0.2 \times 1}{4}$

$T = 40$ dyne/cm $= 4 \times 10^3$ dyne/m

$= \frac{40 \times 10^{-5}}{10^{-2}} = 4 \times 10^{-2}$ N/m

35)

$F_{excess} = (2\pi R) \times 2T$

$= 4\pi RT$

$= 4 \times 3.14 \times 1.5 \times 10^{-2} \times 73 \times 10^{-3}$

$= 1.37 \times 10^{-2}$ N

36) Volume of liquid remain same *i.e.* volume of 1000 small drops will be equal to volume of one big drop

$n\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \Rightarrow 1000r^3 = R^3$

$\Rightarrow R = 10r \therefore \frac{r}{R} = \frac{1}{10}$

$\frac{\text{surface energy of one small drop}}{\text{surface energy of one big drop}} = \frac{4\pi r^2 T}{4\pi R^2 T} = \frac{1}{100}$

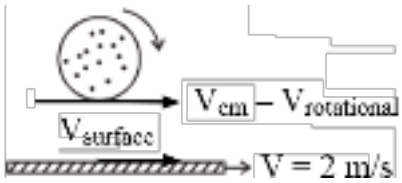
37) Since the particle is moving with constant velocity, let it be at a distance y from the origin. Then its angular momentum will be equal to mvy. Since it is moving parallel to x -axis, therefore y remains constant and in turn the angular momentum (mvr) of the particle remains constant.

38)

$KE \propto L^2$

$$\frac{\Delta L}{L} = \left[\sqrt{\frac{KE_F}{KE_i}} - 1\right] \times 100$$
$$\frac{\Delta L}{L} = \left[\sqrt{\frac{400}{100}} - 1\right] \times 100$$
$$\frac{\Delta L}{L} = 100\%$$

39)



The condition for pure rolling is that the velocity of the point of contact on the rolling body is equal to the velocity of the surface it is rolling on. The velocity of the point of contact is the vector sum of the translational velocity of the centre of mass and the rotational velocity of that point.

$v_{cm} - v_{rotational} = v_{surface}$

$v_{cm} = v_{surface} + v_{rotational}$

$v_{rotational} = \omega R$

$v_{cm} = v_{surface} + \omega R$

Radius of the disc, R = 20 cm = 0.20 m

Angular velocity of the disc, ω = 5 rad/s

Velocity of the moving surface, vsurface = 2 m/s

$v_{cm} = 2 + (5)(0.20)$

$v_{cm} = 2 + 1.0 = 3$ m/s

40) $$I = 0 + 0 + m\left(\frac{\ell}{\sqrt{2}}\right)^2 + m\left(\frac{\ell}{\sqrt{2}}\right)^2$$

$= m\ell^2$

41) $I_{remaining} = I_{complete} - I_{removed}$

$$= \frac{9MR^2}{2} - \left[\frac{M(R/3)^2}{2} + M\left(\frac{2R}{3}\right)^2\right]$$

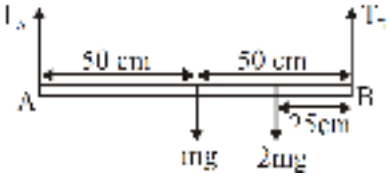
$= 4\ MR^2$

42)

$mg\ \ell_2 = 16\ g\ell_1$ .....(i)

$mg\ \ell_1 = 4g\ell_2$ .....(ii)

$\Rightarrow \frac{16}{m} = \frac{m}{4} \Rightarrow m = 8$ kg



43) $\tau_B = 0$ (torque about point B is zero)

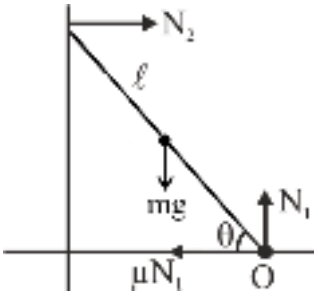
$(T_A) \times 100 - (mg) \times 50 - (2mg) \times 25 = 0$

$100\ T_A = 100\ mg$

$T_A = 1\ mg$

44) For equilibrium

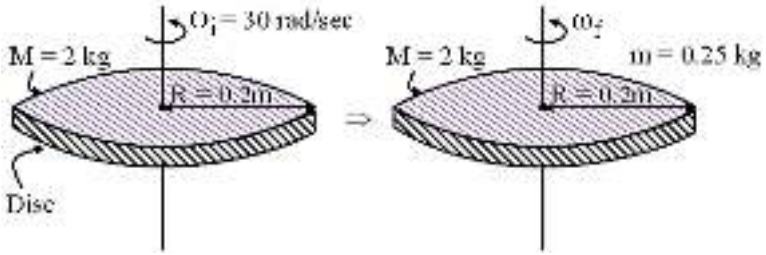
$mg = N_1$



$\mu N_1 = N_2$

Taking moments about 0

$N_2 \cdot \sin\theta = mg\frac{\ell}{2}\cos\theta$



45)

$L = I\omega = \text{constant}$

$L_{system} = \text{constant}$

$I\omega = \text{constant}$

$I_i\omega_i = I_f\omega_f$

$$\left(\frac{mR^2}{2}\right)(\omega_i) = \left(\frac{mR^2}{2} + MR^2\right)(\omega_f)$$

$$\left[\frac{2 \times (0.2)^2}{2}\right][30] = \left[\frac{2 \times (0.2)^2}{2} + 0.25(0.2)^2\right][\omega_f]$$

$0.04 \times 30 = (0.04 + \frac{1}{4} \times 0.04)\ \omega_f$

$1.2 = (0.04 + 0.01)\omega_f$

$1.2 = (0.05)\omega_f$

$$\frac{1.2}{0.05} = \omega_f$$

$$\omega_f = \frac{120}{5}$$

$\omega_f = 24$ rad/sec

46) $\bar{\nu} = \frac{1}{\lambda} = \frac{1}{6000 \times 10^{-10}} = \frac{10^7}{6}$
$= 1.66 \times 10^6\ m^{-1}$

47) For n = 4, ℓ = 0 ; m_ℓ = 0
1 ; –1, 0, 1
2 ; –2, –1, 0, 1, 2
3 ; –3, –2, –1, 0, 1, 2, 3
So for n = 4, $|m_\ell|$ = 1 there are 6 orbitals having 12 electrons out of which electron having $m_s = +\frac{1}{2}$ are 6.

48) $E_n = -13.6\frac{z^2}{n^2}eV$
and $r = a_0\frac{n^2}{z}Å$

49)

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50) $\because 2\pi r = n\lambda$
$2\pi\left(0.529 \times \frac{n^2}{z}\right) = n\lambda$
$\lambda = 2\pi(a_0 \times 3) = 6\pi a_0$

51) $\frac{\frac{1}{\lambda_1} = Rz^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)}{\frac{1}{\lambda_2} = Rz^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right)}$
$\frac{\lambda_2}{\lambda} = \frac{1}{\left(\frac{5}{36}\right)}$
$\lambda_2 = \frac{36}{5}\lambda$

52) $\Delta V = 300 \times \frac{0.001}{100} = 3 \times 10^{-3}$ m/sec
$\Delta x.\Delta V = \frac{h}{4\pi m}$
$\Delta x.(3 \times 10^{-3}) = \frac{0.52 \times 10^{-34}}{9.1 \times 10^{-31}}$
$\Delta x = 1.92 \times 10^{-2} m$

53)

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54) Number of spectral lines = $\frac{n(n-1)}{2}$
$= \frac{6(6-1)}{2} = 15$

55)

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56) NCERT, Pg. # 55

57)

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58) 1st Balmer line n = 3 → n = 2
2nd Balmer line n = 4 → n = 2

$$\Delta E = -13.6z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$$\frac{x = 13.6\left(\frac{1}{4} - \frac{1}{9}\right)}{E_2 = 13.6\left(\frac{1}{4} - \frac{1}{16}\right)}$$

$$\frac{x}{E_2} = \frac{\left(\frac{5}{36}\right)}{\left(\frac{3}{16}\right)} = \frac{5}{36} \times \frac{16}{3}$$

$$\frac{x}{E_2} = \frac{20}{27}$$

$$E_2 = \frac{27}{20}x$$

$E_2 = 1.35\ x$

59)

Z_{eff} in Ar^+ is higher

60)

d-block

61)

Z_{eff} in Na = K (as per slater)

62)

(4) option is exothermic

63)

e^- gain enthalpy order of gp-16

64)

P has max IE

65) Metallic nature P < Si < Be < Mg < Na

66)

Noble gas have ΔH_{eg} = +ve, $_2He \Rightarrow ns^2$

67)

Inner transition metals = f-block

68)

$Be(OH)_2$, $Zn(OH)_2$ (amphoteric)

69)

Both are sp^3

70)

π2p Molecular orbital has nodal plane

71)

$\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2 [\pi(2p_x)^1 = \pi(2p_y)^1]$

72)

SF_4 is see saw

73)

SF_4 is non-planar & polar.

74)

O-nitro phenol contains VWF between molecules

75) BeF_3^- is sp^2 hybrid.

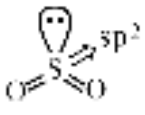
76) If C is alkali metal, A should be halogen (non-metal). Between metal and non-metal ionic bond is present.

77)

'D' is smallest atom

78) 'N' can form NCl_3, N_2O_5 and Ca_3N_2 but not NCl_5.

79) 3pπ – 3pπ bond is unstable

80)  has 1 pπ-pπ & 1 pπ-dπ bonds

81) Solid form of PF_5 is remains same

Solid form of PCl_5 is $PCl_4^+\ PCl_6^-$
$sp^3\ \ sp^3d^2$

Solid form of PBr_5 is $PBr_4^+Br^-$
sp^3

Solid form of XeF_6 is $XeF_5^+F^-$
$sp^3d^2\ \ sp^3d^4$

82) In O_2 molecule, the energy of σ_{2pz} is less than π_{2px} and π_{2py}.

83) **Question Explanation:** Identify the incorrect order regarding covalent character.

Concept: Fajan's Rule and Cation Charge.

Solution: Higher charge on the cation increases its polarizing power and thus increases the covalent character of the compound. Therefore, covalent character should increase

from NaCl to $AlCl_3$.

A. **Trend 1:** Covalent character NaCl > $MgCl_2$ > $AlCl_3$ is **incorrect**.
B. **Correct Trend:** $AlCl_3$ > $MgCl_2$ > NaCl.

Final Answer: Option (1)

84)

BeO has greater LE

85)

NaCl is ionic

86)

Intermolecular forces

87)

$Na_{(s)}$ is metallic solid

88)

Reason is correct explanation.

89)

	NO_2^+	NO_2	NO_2^-
	(sp)	(sp^2)	(sp^3)
ℓ_p	0	0.5	1

90)

B_2 contain only σ bond in absence of s-p mixing

BIOLOGY

91)

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92)

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93)

NCERT-XI, Pg. # 96

94) NCERT Pg. No. 88

95) NCERT Pg. No. 93

96) NCERT-XI, Pg. # 94

97) NCERT-XI, Pg. # 91

98) NCERT-XI, Pg. # 100

99) NCERT-XI, Pg. # 99

100) NCERT-XII, Pg. # 100

101) NCERT-XI, Pg. # 126

102) NCERT-XI, Pg. # 126

103) NCERT Page No. # 162

104)

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105)

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106)

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107) NCERT XI Pg. # 109

108)

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109)

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110) NCERT-XII, Pg. # 118

111) NCERT-XI, Pg. # 117

112) NCERT-XI, Pg. # 116, 117

113) NCERT Pg. # 116

114) NCERT-XI, Pg. # 110

115)

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116) NCERT-XI, Pg. # 118

117)

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118) NCERT Pg. # 108

119) NCERT Pg.# English- 95,96,98,100.
Hindi- 96,97,98,100.

120) NCERT-XI, Pg. No. # 99, 100

121) NCERT Page No. 89, 90

122) NCERT Pg. # 123

123) NCERT, Pg. # 121

124) NCERT-XI, Pg. # 121

125) NCERT XII Page No. # 123 [New]

126) NCERT XII Page No. # 126 & 127 [New]

127) NCERT-XI Pg. # 123

128) NCERT (XI) Pg. # 122,123,124

129)

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130) NCERT-XI, Pg. No. # 87

131) NCERT Pg.#96, 97.

132) NCERT Pg. No. 93, 94

133) NCERT-XI, Pg. # 97, 98, 99 & 100

134) NCERT, Pg # 89, 90, 91

135) NCERT Pg.#100,101,102.

136) NCERT-XI, Pg. # 198

137)

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138)

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139)

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140)

NCERT XI Pg # 285

141) NCERT XI Pg. # 286

142)

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145)

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147) NCERT-XI, Pg. # 200

148) NCERT-XI, Pg. # 203

149)

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150)

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151)

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152)

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153)

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154)

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155) NCERT Pg. # 188

156) NCERT, Pg # 274

157) NCERT, Pg. # 190

158) NCERT, Pg. # 190

159) NCERT, Pg # 190

160) NCERT-XII, Pg. # 276

161)

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162) NCERT Pg. # 268

163)

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164)

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165) NCERT, Pg # 183,184

166)

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167) NCERT-XI, Pg. # 184-185

168)

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169)

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170) NCERT-XII, Pg. # 297

171)

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172)

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174)

Module

175)

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176)

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177)

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180)

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